Workday Accounting Journal EIB Ingest Studio

DESKTOP EDITION

Standalone Enterprise Desktop Application (.exe) — Comprehensive User Guide

Executive Summary: This guide explains in simple, plain English how the standalone `WorkdayEIBStudio.exe` desktop application works, what the transformation engine `run_agent.py` does behind the scenes, and provides a clear, step-by-step walkthrough on how to upload raw accounting files, choose save locations, select templates, and generate balanced Workday spreadsheets.

1. What Is This App & How Does The .EXE Work?

Workday EIB Ingest Studio is a standalone Windows desktop software (`WorkdayEIBStudio.exe`) designed to solve a critical corporate accounting problem: converting raw accounting journal data (from CSV or Excel extracts) into the exact, formatted spreadsheet templates required by **Workday Enterprise Interface Builder (EIB)**.

STEP 1: INPUT	STEP 2: PROCESSING	STEP 3: OUTPUT
Raw Accounting Data	**ETL & Validation Engine**	**Ready-to-Upload EIB**
Source journal files with raw lines, amounts, cost centers, and memos.	Maps accounts, formats cost centers, checks balance, cleans NULL memos.	Official Workday workbook perfectly balanced to $0.00 variance.

How Does The Executable Work on Anyone's Computer?

- **Zero Installation Required:** It is a single, self-contained file (`WorkdayEIBStudio.exe`). The user simply double-clicks it to run.
- **No Python or Libraries Needed:** Python, pandas, openpyxl, and all templates are bundled inside the executable. The recipient does not need to install anything.
- **Pure Native Desktop (No Web Server / No Open Ports):** Unlike web apps, it opens a native Windows window using Microsoft Edge WebView2 and communicates via direct in-memory calls. It does not open port 8080 or network sockets, meaning **Windows Defender Firewall will never trigger a warning**.
- **Embedded Workday Templates:** All official Workday EIB base templates (v45.1, v46.0, v47.0, v48.0) and the Spend Category Mapping dictionary are embedded inside the `.exe` file itself.

2. The Desktop User Interface Overview

The interface is organized into a clean, distraction-free desktop utility workbench:

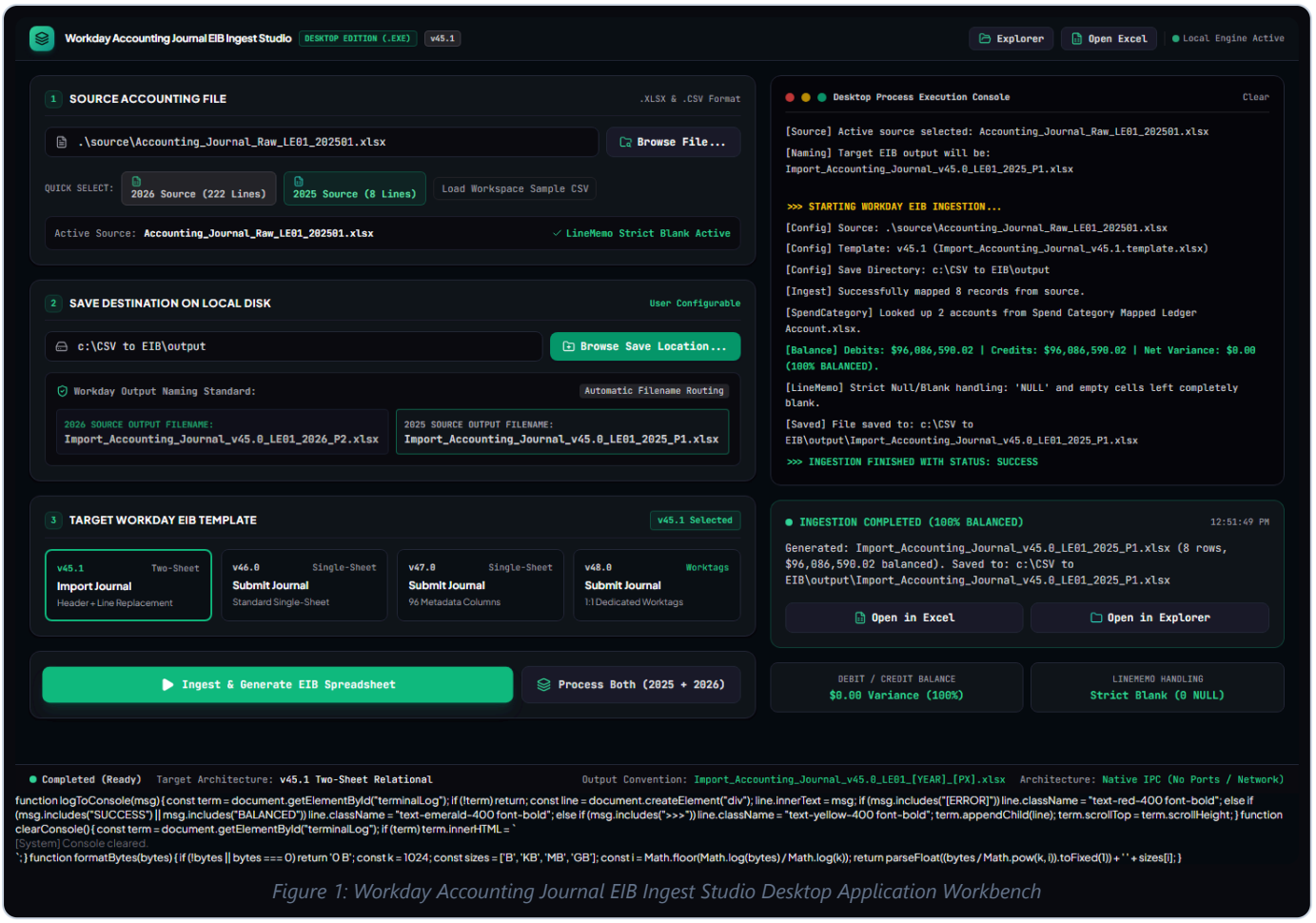


Figure 1: Workday Accounting Journal EIB Ingest Studio Desktop Application Workbench

3. Step-by-Step User Walkthrough (Layman Terms)

Here is the exact step-by-step procedure to convert any raw accounting file into a balanced Workday EIB spreadsheet:

1 Launch the Application

Double-click `WorkdayEIBStudio.exe` on your Windows desktop or file explorer. The desktop studio window will open immediately. You will see the top titlebar display **"DESKTOP EDITION (.EXE)"** and **"Local Engine Active"**.

2 Select Your Source Accounting File (Panel 1)

You have three easy ways to select your raw file:

- **Browse File... button:** Click this button to open the standard Windows Open File Dialog and choose any `.xlsx` or `.csv` file from your computer or network drive.
- **Quick-Select Pills:** Click **"2026 Source (222 Lines)"** or **"2025 Source (8 Lines)"** to instantly load the workspace source files.
- **Drag & Drop:** The app automatically verifies file existence and updates the active source indicator.

3 Choose Where to Save the Output (Panel 2)

Decide where you want the final generated Workday spreadsheet to be saved on your computer:

- By default, it will save to an `output\` folder located in the same directory as the app.
- To change this, click **"Browse Save Location..."**. A native Windows folder picker will appear. Select any folder (such as your `Desktop`, `Documents`, or a shared drive like `S:\Accounting\Workday\`).
- Notice the **Workday Output Naming Standard** cards below the path. The app automatically routes your output name:
  • 2025 source → `Import_Accounting_Journal_v45.0_LE01_2025_P1.xlsx`
  • 2026 source → `Import_Accounting_Journal_v45.0_LE01_2026_P2.xlsx`

4 Select Your Target Workday EIB Template (Panel 3)

Click on one of the 4 visual template cards depending on your company's Workday setup:

- **v45.1 (Two-Sheet Relational — Default):** Generates both *Import Accounting Journal* (headers) and *Journal Entry Line Replacement* (lines).
- **v46.0 (Single-Sheet):** Standard Workday *Submit Accounting Journal* single-sheet structure.
- **v47.0 (Single-Sheet Worktags+):** Extended 96-column Workday template format.
- **v48.0 (Dedicated Worktags):** Latest Workday architecture with 1:1 dedicated columns for Cost Center, Spend Category, BU, etc.

5 Execute Ingestion & Generation (Panel 4)

Click the primary green button: **[Ingest & Generate EIB Spreadsheet]**.
*(Alternatively, if you want to process both 2025 and 2026 files simultaneously, click **[Process Both (2025 + 2026)***

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6 Verify Live Console Logs & Status (Panel 5 & 6)

Watch the dark monospace execution terminal on the right. In less than 1.5 seconds, it will:

- Log how many source records were read (e.g. 222 rows for 2026, 8 rows for 2025).
- Report how many Spend Categories were mapped from the lookup table (e.g. 56 accounts).
- Verify debit sum vs. credit sum: confirms **"Net Variance: $0.00 (100% BALANCED)"**.
- Confirm **"LineMemo Strict Null/Blank handling: 'NULL' and empty cells left completely blank"**.
- Display the green status badge: **INGESTION COMPLETED (100% BALANCED)**.

7 Open & Upload to Workday

You are now ready to inspect and upload your spreadsheet:

- Click **[Open in Excel]** to instantly open the generated spreadsheet in Microsoft Excel.
- Click **[Open in Explorer]** to highlight the output file in Windows File Explorer.
- Log into Workday, navigate to the **Integration: Inbound EIB** task, and upload the generated `.xlsx` file.

4. Under the Hood: What Does `run_agent.py` Do?

`run_agent.py` is the **Extraction, Transformation, and Loading (ETL) engine** of the application. It contains all the financial rules, lookup dictionaries, data validation routines, and Excel generators.

Key Operations Performed by `run_agent.py`:

Operation	Business Rule	How run_agent.py Implements It
LineMemo Strict Blank Rule	If a memo is empty or contains `"NULL"`, it must remain strictly empty in Workday with no fallback text.	The `clean_memo()` function checks for `NULL`, `NONE`, `NAN`, or whitespace and returns Python `None`. In openpyxl, this writes a true blank cell with zero characters.
Spend Category Lookup	Workday requires a Spend Category for ledger accounts.	`load_spend_category_map()` loads 108 mapping pairs from `Spend Category Mapped Ledger Account.xlsx`. During line mapping, it matches account codes and injects the corresponding spend category.
Cost Center Standardization	Cost center codes must be standard 3-digit identifiers.	`clean_cost_center()` takes numbers like `0` or `90` and formats them with leading zeros as `'000'` and `'090'`, applying Excel text format (`@`).
Date Formatting	Workday accounting dates must follow ISO format without time stamps.	`parse_date()` converts date strings to true Python `datetime.date` objects, preventing Excel serial number display bugs.
Amount Cleanliness & Balance	Debit and credit amounts must equal each other with $0.00 difference.	`clean_amount()` parses amounts to 2 decimals, preserves zero amounts (`0.0`), calculates total debits and credits, and asserts `round(debits - credits, 2) == 0.00`.
Relational Two-Sheet Structure (v45.1)	Headers and line items must link via `Header Key*`.	• **2026:** Groups unique journals by JournalKey/AccountingDate into 20 Header rows (Sheet 1) and writes 222 Line rows (Sheet 2) linked via Header Keys 1 through 20. • **2025:** Writes 1 Header row (Header Key 1) and links all 8 Line rows to Header Key 1.
Single-Sheet Structure (v46, v47, v48)	For newer Workday templates, headers and lines sit on the same line row.	`process_journal_single_sheet()` writes line data and places header fields (Accounting Date, Company, Ledger) directly on row 6 onwards, mapping dedicated worktags for v48.0.
File Naming Convention	Output files must strictly reflect company period conventions.	`get_output_filename()` automatically assigns: • 2025 → `Import_Accounting_Journal_v45.0_LE01_2025_P1.xlsx` • 2026 → `Import_Accounting_Journal_v45.0_LE01_2026_P2.xlsx`

5. Troubleshooting & FAQs

Q: What if I see "File is currently open in Microsoft Excel"?
A: Windows locks Excel files while they are open. If you previously generated a file and have it open in Excel, simply close Excel and click **[Ingest & Generate EIB Spreadsheet]** again.

Q: Can I process CSV files instead of Excel files?
A: Yes! The source file selector accepts both `.xlsx` and `.csv` formats. `load_source_dataframe()` automatically detects the file extension and loads it correctly.

Q: Do I need internet access to run the app?
A: No. The application runs 100% locally on your computer. All processing, calculations, and file generations happen entirely offline without sending any data over the internet.